

1a) The main objectives of computer vision are

- To acquire images.
- To analyze and understand visual data
- To recognize objects
- Helps for decision making.

b) Real-World Applications are

- Face-recognition
- self-driving cars.

1. The computer vision enables computers to interpret and understand visual images by processing text or the pixels into images. Further the images are broken into pixels that are colourful or Black and white

2. The key features of computer vision that make it suitable for intelligent image analysis are:

- Feature extraction
- Object recognition
- Image understanding and
- Pattern Recognition

a) Feature extraction

It is the process of identifying the

important characteristics such as edges, corners or shapes from an image.

b) Object recognition improves the Computer vision by its accuracy and precision to identify and classify objects in images or videos.

3. Levels of image processing.

There are three levels of image processing such as -

- Low
- Mid
- High

Low level

Enhances quality and helps in noise reduction

Mid level

Feature Extraction and Object recognition

High Level

Image understanding and Pattern Recognition.

a) Low-Level Image processing

The Low-Level Image processing is the process where the images are generated at a lower size or quality. It usually enhances quality for this particular level and helps for noise reduction.

b) The difference between mid-level and high level image processing

In a mid level image processing the images

are processed in a medium quality or size. The mid level image processing mainly focusses on Feature extraction and Object recognition. In a high level image processing, the images are processed in the highest quality or size possible. The high level image processing mainly focuses on Image understanding and pattern recognition.

4. Digital image.

- The energy is converted into electrical in order to generate a digital image.
- It shows a real world scene in a digital ~~form~~ ^{form} stored, interpreted and processed by a digit computer.

Fundamental Components.

→ Pixels

→ Images.

It is defined as a 2D function $f(x, y)$

a) A pixel is the smallest unit of that is present in a pixel. It is a colourful or Black and white detail that is small in size to generate images.

b) Image resolution affects image quality by totally destroying the output of the image. Eventhough the image is good enough to visualize the resolution decides the quality of an image.

the clear picture can be identified if we balance both equally.

5. Conversion takes place through the following:

- * Scene Illumination
- * Reflection or Transmission
- * Sensor Capture
- * Analog to Digital Conversion.

a) Role of an image sensor.

It converts light into electrical signal for digital image creation.

Here, light refers to energy.

b) 2 image acquisition devices.

- * Digital Camera
- * Scanner.

6. Formation of digital image.

The real world images captured is continuous in both space and intensity. To store and ~~convert~~ process it on a computer, it must be converted into a digital image using sampling and quantization.

a) Sampling

- Process of converting a continuous image in space into a discrete set of pixels
- Determines how many pixels represent the images

6) Purpose of Quantization:

Reduces image size.

Image Compression

Speeds up Processing

Simplifies Image Analysis

Controls Image Quality

7. Image formation model describes the process of capturing images as how light from objects passes through a camera lens and forms an image on the sensor.

a) Perspective projection converts a 3D scene into a 2D image while preserving depth

b) Illumination provides light, makes objects visible for image capturing.

8. Image processing enhances images, while computer vision interprets the images and makes decisions

a) Image processing enhances images, analyzing or modifying digital images.

b) Image Processing
Enhance images

Output is improved image

Computer Vision

Understands images

Output is information or decision

9. Computer vision is used in Healthcare, Agriculture, Finance etc in the real world problem solving.

a) Medical image analysis is used to detect diseases such as tumors or fractures from X-rays, CT scans and MRI images.

b) Computer vision helps autonomous vehicles to detect lanes, traffic signs, pedestrians and obstacles for safe navigation.

10. Digital images fundamentally are pixels, resolution, sampling and quantization provides the data required for image analysis.

a) Because every digital image is made up of tiny pixels that store color and intensity information.

b) Higher sampling and resolution capture more details, improving the accuracy of image analysis.